Water Team Description 2026

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Abstract. This paper focuses on Team Water's robotic systems, highlighting key advancements in robot vision, software architecture, path planning, self-positioning, and collaborative strategies. It summarizes the team's RoboCup and other prestigious platform achievements, including YOLOv11 integration for football recognition, Middle Size League master selection algorithm optimization, and inter-system communication-driven performance enhancement—with middle-sized and humanoid robots competing together. It also addresses vision detection program version compatibility issues and proposes an innovative solution ensuring existing OS compatibility while adapting to new demands.

Keywords: Robocup Middle Size, Water, Humanoid robot, Deep learning, Embodied intelligence

1 Introduction

The Beijing Information Science & Technology University proudly presents Team Water, its distinguished Middle Size Robot Team, established in 2003. Renowned for their participation in the prestigious China Open from 2006 to 2025, and their commendable appearances in the World Championships from 2010 to 2019, the team exemplifies excellence in robotic innovation. Since 2025, Team Water has also begun participating in the Humanoid Robot League, boasting corresponding technologies as well as eligible team members and robots for the competition. Team Water's expertise is demonstrated through their in-depth research in cutting-edge domains, including Robot Vision, advanced Software Architecture Frameworks, intricate Path Planning mechanisms, precision-oriented Positioning techniques, sophisticated Communication forms, and robust Control Models, placing them at the forefront of technological advancement in the field of robotics.
This paper focuses on Team Water's robotic systems, highlighting key advancements in robot vision, software architecture, path planning, self-positioning, and collaborative strategies. It summarizes the team's RoboCup and other prestigious platform achievements, including YOLOv11 integration for football recognition, Middle Size League master selection algorithm optimization, and inter-system communication-driven performance enhancement—with middle-sized and humanoid robots competing together. It also addresses vision detection program version compatibility issues and proposes an innovative solution ensuring existing OS compatibility while adapting to new demands.

World Championships: 1st place: 2010,2011,2013,2015,2017 2nd place:2014,2019
3rd place: 2013,2018

RoboCup China Open: 1st place: 2010,2014,2017,2018,2019，2021，2023，2025
2nd place: 2008,2009,2011,2013,2016,2024
Top 1：2022

Asia Pacific RobotCup: 1st place: 2019，2021，2023
3st plase:2022

Capability video： https://youtu.be/HqMzqoP1ENk

2 Hardware Structure

The original robot was made manually in 2004.The first two wheels robot was made manually

by the team in 2006.Then in 2007 we made a large-scale transformation to it, and advanced the visual parts and motor driver parts of the robot. So the effect of the image and the speed of robot were upgraded.(**Fig. 1 and 2**)In 2008, our first Omni-directional three wheels robot was finished which was also made manually.(**Fig.3**)This robot was more flexible. In 2009, our robot was advanced and new robot was born.

Now we are using the latest type of robot.(**Fig.4**)

Dimensions: 50*50*80(cm)
Weight: 45kg(with one battery inside)

Fig.1 Fig.2 Fig.3 Fig.4

2.1 Motion

When we made the robot, we put an Omni-directional wheel, a 160w DC electric machinery and a pro-Motion BDMC3606SH Servo Driver together to make a Machinery Servo Driver. And placed them on the evenly Equilateral triangle robot machine chassis. The angle of the adjacent otor axis is 120 degrees. So the connection between the Machinery servo Driver is simpler and easier to replace, remove or fix.(**Fig.4 and 5**)

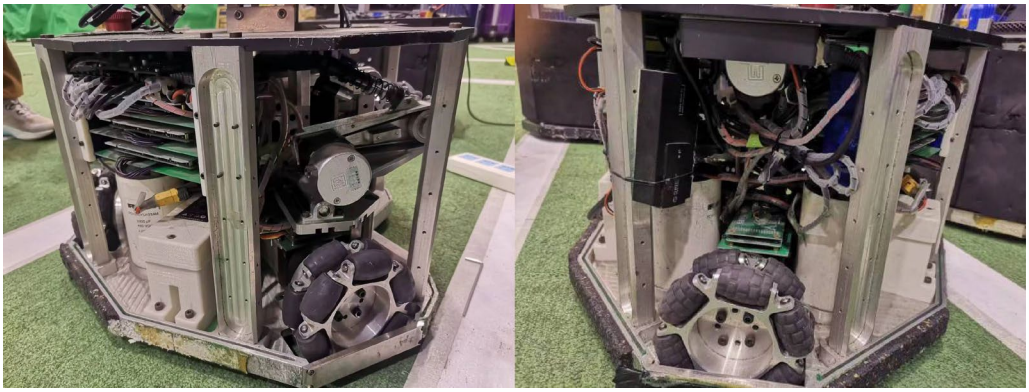

Fig.5

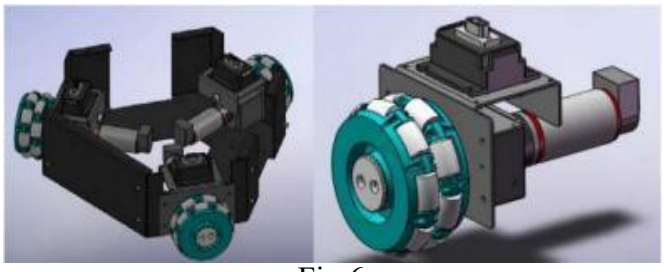

Fig.6

2.2 Control circuit

Our control circuit is capable of uniformly controlling all functions of the robot. The onboard computer can communicate with the control circuit via a USB interface. Through this interface, we can control the robot's 5-axis servo motors and collect data including the current of each motor, encoder positions, and the dribbling volume from the dribbling sensor.

2.3 Camera

The minimum requirement for cameras is 640*480 resolutions and 45 fps. Therefore we select the point Gray's fumy camera, which can reach 60 furthest color dynamic range was higher than 90db, so we had a better suitability for light.(**Fig.7**)

By the way, We tested a panoramic vision system composed of six 90-degree RGB cameras arranged in a circular array. Compared to panoramic imaging systems using single cameras with reflective mirrors, this configuration exhibits significantly reduced distortion, offers broader application scenarios, and demonstrates greater potential for performance enhancement and adaptability.(**Fig.8**) The system is currently in the experimental phase.

2.4 Omni

The visual range of omni line and the ball we made was much more than 9m. It made it possible that if there was no shelter materials in the ring, our robot would overlook the court nearly about more than 90%.This character greatly helped our robot to judge the trace and predict the position of ball.

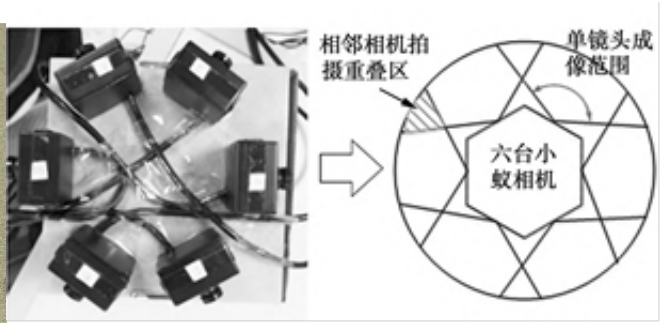

Fig.7 Fig.8

3 Software System

3.1 Path-Planning

Our path-planning system can not only provide the optimal walking route but also enable the robot to proactively generate its own actions. Based on its direction in the polar coordinate system and other relevant models, the robot determines its movement direction, speed, and angular speed, thereby forming its unique movement mode. This feature offers several advantages: it cleverly integrates the tactical and control components, generating a unified solution for movement actions. Meanwhile, the robot can flexibly judge strategies (such as whether to attack or defend) based on its own position as well as the positions of teammates and the ball. For instance, in defense scenarios, the coach only needs to send a simple command to assign the robot to a specific location, and the robot will independently generate its movement pattern. This provides a simple, effective, and flexible approach to motion control.

Due to this feature, we also could set up any of tactics flexibility. Our robot would do technical adjustments for different team, strategy and formation. We often did some special adjustment in the second halfjust for the problem in the first half.

3.2 Self positioning

Our robot self-locates by the white lines of the field relative to the position and angel of the robot. We will make correction to the omni-image first and recovery the shape of field line.(**Fig. 9**)The picture is Omni-image before correction and after correction. Through the white line in the image and template of field line, we get the location relative to the field line. From **Fig.10** , the color of template field line is blue. The white field line was expressed by pink matrix. The software will match the pink matrix with the blue lines until they get together. We calculate absolute coordinate in the field with the turning of matrix angel and distance.

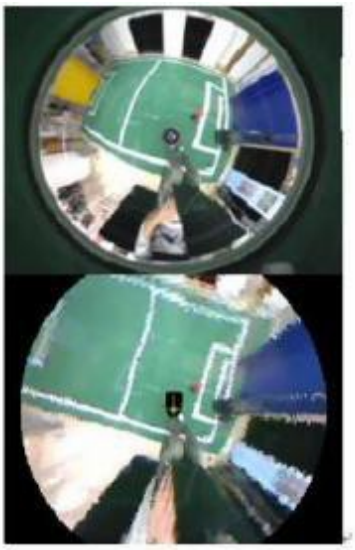

Fig.9

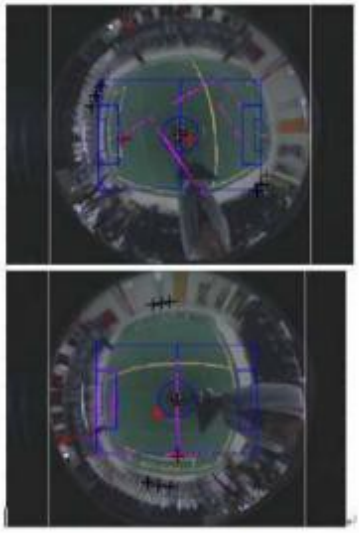

Fig.10

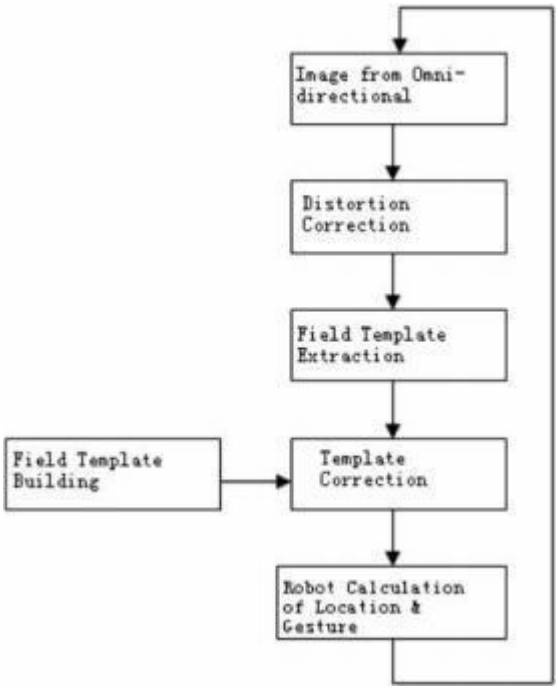

Fig.11

4 Pioneering Ingenuity

Over the past year, our team has made substantial strides in the field of robotics technology, achieving encouraging progress. As researchers and learners, we have consistently maintained a strong desire for knowledge and a pursuit of innovation. This year, we concentrated on three pivotal projects, which not only enhanced our practical experience but also profoundly impacted our understanding and application capabilities in technology.

Firstly, we endeavored to implement the cutting-edge YOLOv11 technology in our football recognition system. This effort not only advanced our skills in object detection but also ignited our passion for real-time image processing techniques. Secondly, we meticulously refined the master selection algorithm used in the RoboCup Middle Size League (MSL) football robot competition. This process not only honed our algorithm design skills but also deepened our comprehension of team collaboration and strategic planning. Finally, by realizing inter-system communication technology on our Water football robots, we not only broadened our understanding of robotic communication networks but also laid a solid foundation for future multi-system integration projects.

4.1 Football Robot Following System Based on YOLOv11

In alignment with the evolution of the RoboCup Middle Size League (MSL) competition towards human-robot confrontation modes, we proposed a diverse football recognition solution based on the YOLOv11 model. This system integrates the principles of binocular ranging for spatial localization, enabling the robot to follow the movement of the

football. The overall software system was initially tested on the Jetson Xavier NX embedded platform. Experimental results indicate that the visual recognition operates at approximately 56 frames per second, fulfilling the real-time requirements of the competition. The recognition accuracy in a complex environmental validation set reached 0.767mAP@0.5, meeting the precision demands of thecompetition.

The vision system approach of the RoboCup MSL has transitioned from simple color-based recognition methods to advanced recognition algorithms based on deep learning, offering multiple possibilities for enhancing robot performance in complex environments. Building on this, our team developed a football recognition system based on the YOLOv11 deep learning network. This system, integrated with depth information provided by depth sensors, localizes the football and guides the robot to track it. The system uniquely identifies and locates footballs of various colors and patterns, addressing the gap in robot recognition when the football is airborne.

The vision recognition and localization node detects the football's position in real-time and communicates with the robot's motion control node to execute the movement control of the football robot. The control outcomes are illustrated in **Fig.11**. The experimental procedure involved kicking the football along the midline of the field, with the robot's tracking performance observed from a third-person perspective. Image (1) depicts the initial state, and in Image (2), it is evident that upon receiving the football's position, the robot calculates the direction, adjusts its posture, and aligns itself towards the football. Images (3) to (8) display the continuous movement of the football, with the robot promptly adjusting its direction and ultimately reaching the position to kick the football, as shown in Image (9). **Fig.12** presents the first-person perspective images corresponding to **Fig.13** during the experiment, illustrating the robot's consistent detection of the football (Image 12 captures the robot's perspective when it has seized the ball, with other sensory methods determining that it's in a ball possession state).

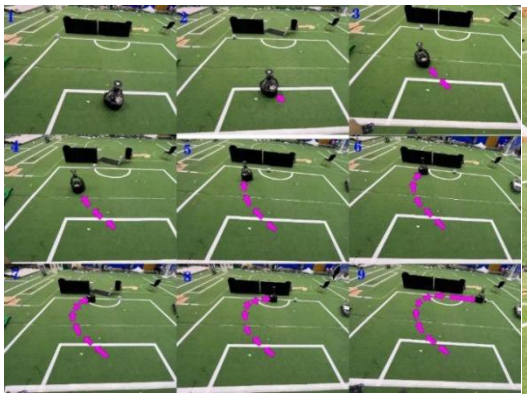

Fig.12 Robot Follows Soccer Ball Movement Trajectory

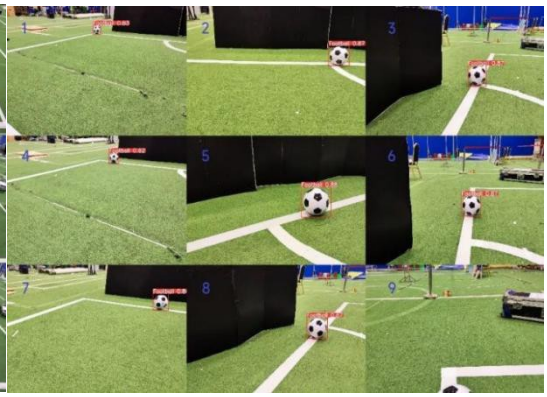

Fig.13 Robot Vision Recognition

To validate that this vision solution can complement the robot's recognition of airborne footballs, the following experiment was designed: the football was placed in a stationary position on the ground, and the recognition system was activated. The ball was then slowly elevated from the ground in the vertical direction, as depicted in **Fig.14**. The direction of the ball's ascent was consistent with the negative direction of the y-axis in the camera's coordinate system. The football positions obtained by the recognition and localization algorithm are presented in **Fig.15**.

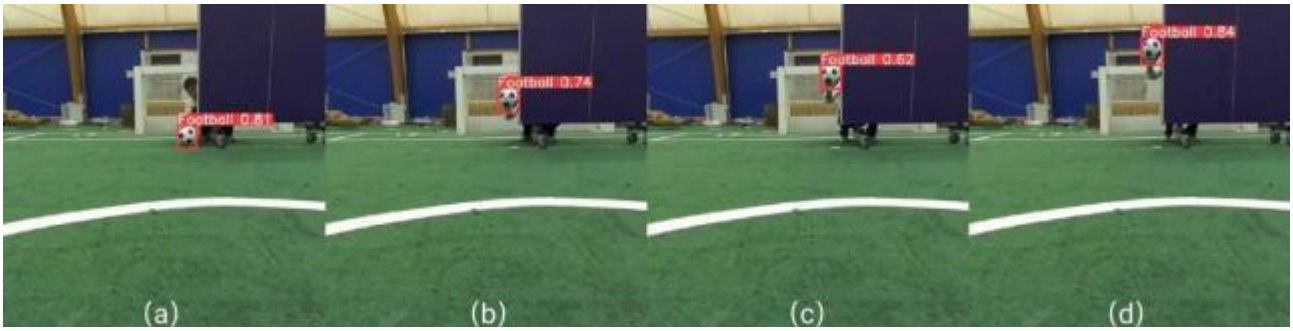

Fig.14 Football Lifting Process

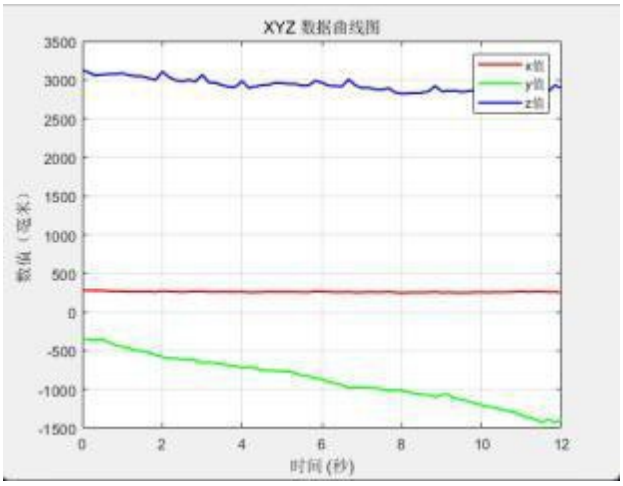

Fig.15 Positioning Results of the Ball Lifting Process

As illustrated in the figure, the football is gradually elevated from (a) to (d), during which the corresponding y-values continuously decrease, while the x and z-values remain relatively constant. This is in accordance with the motion trajectory of the football, thus validating that the system is capable of accomplishing the task of recognizing airborne footballs.

Through analysis and comparison, the YOLOv11 model is selected as the object detection algorithm and further improved. The improvements mainly focus on enhancing detection accuracy and lightweighting the model, making it suitable for the on-board platform of soccer robots with limited computing power.

A deformable large-kernel attention module is designed by integrating deformable convolutional neural networks and a global attention mechanism into YOLOv11. This module can capture richer feature information, strengthen the model's attention to target features, and significantly improve the multi-scale object detection performance of the model.

Based on the convolutional block attention module (CBAM), a lightweight EAM model is proposed through optimization. This attention model introduces only a small number of parameters while achieving considerable performance improvement.

For the panoramic vision system, we have implemented an image matching and stitching system. Feature points and descriptors are extracted from the images using the ORB algorithm. A brute-force matcher is then employed to establish initial correspondences, followed by the selection of high-quality matching points. The images are aligned and stitched through perspective and affine transformations, with the final mosaic result being both saved and displayed. Additionally, the SIFT algorithm is integrated for keypoint detection and local invariant feature extraction, with RANSAC utilized for robust feature matching refinement. The system is structured around a dedicated Stitcher class, which encapsulates the complete image stitching pipeline. Leveraging the OpenCV library for image and video processing, the program continuously reads frames from two video streams, performs real-time stitching, and outputs the composite result. The feature matching between the two images can be visualized by enabling the showMatchesparameter. The system operates in a continuous loop, processing video frames sequentially to generate the stitched output.(**Fig.16**)

Fig.16 Panoramic image stitching effect and recognition results

4.2 Technical Report on the Implementation of Inter-system Communication in Water Football Robots

The primary objective of this study is to address the issue of version compatibility in the development of vision detection programs for football robots. Control programs for football robots typically require operation in specific system environments, such as running a VC6.0- based football robot control program on an older Windows XP system. However, with ongoing iterations of the robot, rapid development of new functionalities is necessary to meet the requirements of different competition or working environments. Often, this rapid development involves the use of off-the-shelf integrated algorithm programs, such as the OpenCV Python package for vision processing. However, many powerful algorithm integration programs are not compatible with relatively outdated operating systems, making integration into existing robot programs challenging.

When faced with the aforementioned issues, one solution is to comprehensively replace the robot program to adapt to newer versions of the operating system and to be compatible with new integrated programs. However, this approach is not always feasible. Especially when the robot control program is highly complex and contains a significant amount of algorithm code dependent on the current operating system environment, replacing new algorithm code typically requires considerable time to resolve compatibility issues with the new system. Furthermore, although this method is more effective and standardized in the long

Subsequently, the processed vision data information is returned to the XP system through the message queue method.

The implementation process is shown in the figure：

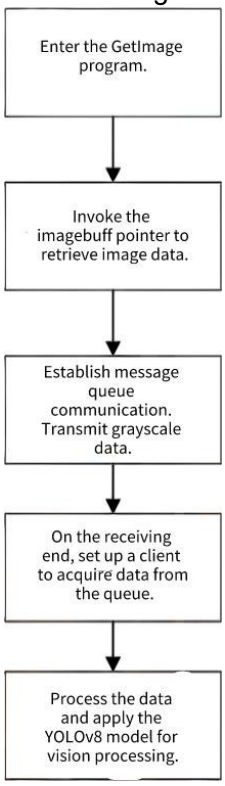

Fig.17 Program implementation basic flow

In the Windows XP system, specifically within the subordinate machine of the football robot, a message queue is established, and the byte-type array of vision data pointed to by the buffer pointer is transmitted. The critical code for this part is illustrated in the figure.

```
std::vector<char> buffer((std::istreambuf_iterator<char>(file)), std::istreambuf_iterator<char>());
auto_ptr<BytesMessage> message(session->createBytesMessage());
message->setBodyBytes(reinterpret_cast<unsigned char*>(buffer.data()), buffer.size());
producer->send(message.get());
```

Fig.18 The server sends visual data

After retrieving and transmitting the vision data from the main program of the Water robot in the subordinate machine system, the vision data is acquired and processed through deep learning on the Win10 operating system. The key code for the data reception portion is depicted in the accompanying figure.

```
std::vector<unsigned char> imageBytes;
int length = bytesMessage->getBodyLength();
imageBytes.resize(length);
bytesMessage->readBytes(&imageBytes[0], length);
processImage(imageBytes);
std::cout << "Received and processed image data." << std::endl;
```

Fig.19 The client receives a visual data array

After addressing issues related to continuous dynamics and data

overwriting, the data transmission is completed, along with the vision processing based on the YOLOv11 model.

4.3 Humanoid robot

Currently, Team Water possesses not only the capability to compete in the Middle Size League but also fields a humanoid robot group eligible for participation. The team is equipped to engage in both 3 vs. 3 and 5 vs. 5 competitions, utilizing Booster T1 soccer robots. With extensive experience in domestic competitions in China, the team has achieved notable results. Furthermore, the team employs reinforcement learning techniques to optimize robot gait training for kicking motions and to enhance autonomous decision-making capabilities.

In terms of hardware, we possess the necessary infrastructure to support simultaneous participation and on-field competition in both the Middle Size League and the Humanoid Robot League.(**Fig.20**)

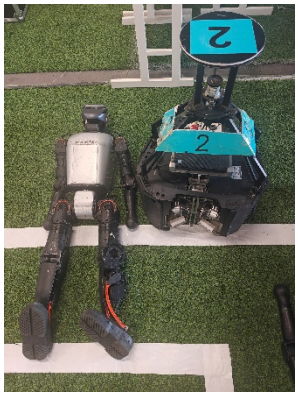

Fig.20 Booster T1 robot

This robot measures 118 cm in height, 47 cm in width, and 23 cm in thickness. It features arm lengths of 45 cm and leg lengths of 57 cm, with an approximate weight of 30 kg. The robot possesses a total of 23 degrees of freedom (DoF), distributed as follows: six DoF per leg, one DoF at the waist, four DoF per arm, and two DoF in the head. In terms of computational and sensory capabilities, it is equipped with an NVIDIA AGX Orin platform delivering 200 TOPS of AI computing power, along with a depth camera and a 9-axis inertial measurement unit (IMU).

Till now, we have 6 robots like this, except competition, we also train these robots to participate in performance.

5 Summary

The study encapsulates the comprehensive development of Team Water's robotic systems, detailing the hardware structures, control circuits, and camera configurations that form the backbone of both its middle-sized and humanoid robots. It delves into the software system, elaborating on the sophisticated path-planning, self-positioning, and vision processing functionalities that empower both types of robots with advanced tactical and strategic capabilities. Notably, the research reflects Team Water's dedicated efforts in transitioning from the Middle Size League to the Humanoid Robot League, as the team strives to tackle and accomplish tasks that have not been achieved before in this field. The core of the research lies in the pioneering ingenuity applied to the football recognition system, leveraging YOLOv11 technology and robust inter-system communication to navigate complex environments and achieve real-time responsiveness—technologies that are also being adapted and optimized for humanoid robots to bridge the technical gap between the two leagues. The paper addresses the crucial aspect of version compatibility in software development, proposing a practical solution that ensures the swift implementation of new features without compromising the existing system structure, which is equally applicable to both middle-sized and humanoid robotic systems. Through this exploration, Team Water continues to push the boundaries of robotic technology, aiming for excellence in future RoboCup and other robotic competitions; moreover, the team is committed to sharing its research achievements, practical experience, and problem-solving methods with the academic and industrial communities to promote the collective progress of robotic technology in both leagues.

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